TD n° 9 (graded) Hugo SACCU

Exercise 1.

We define

$$A \tilde{\wedge} B \quad \text{by} \quad (A \Rightarrow B \Rightarrow \bot) \Rightarrow \bot$$

and $A \tilde{\vee} B$ by $(A \Rightarrow \bot) \Rightarrow (B \Rightarrow \bot) \Rightarrow \bot$.

They are equivalent by $(A \Rightarrow B \Rightarrow C) \Leftrightarrow (A \wedge B \Rightarrow C)$
and $(A \Rightarrow \bot) \Leftrightarrow \neg A$

to $\neg\neg(A \wedge B)$ and $\neg(\neg A \wedge \neg B)$
which are equivalent in NC (but not NJ) to

$$A \wedge B \quad \text{and} \quad A \vee B$$

respectively.

If, in NJ, one shows the equivalence

$$\vdash (A \tilde{\wedge} B) \Leftrightarrow (A \wedge B) \qquad \text{for all } A, B$$

then we know $\vdash \neg\neg A \Rightarrow A$ in NJ by

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash \neg\neg A}\,ax \qquad \dfrac{\dfrac{\dfrac{}{\Gamma, A \vdash A}\,ax \qquad \dfrac{}{\Gamma, A \vdash \top}\,\top_i}{\Gamma, A \vdash \bot}\,\Rightarrow_E + ax}{\Gamma \vdash \neg A}\,\Rightarrow_i}{\Gamma := \neg\neg A,\ (A \Rightarrow \top \Rightarrow \bot) \vdash \bot}\,\neg_e}{\neg\neg A \vdash A \tilde{\wedge} \top}\,\Rightarrow_i}{\dfrac{\neg\neg A \vdash A \wedge \top}{\neg\neg A \vdash A}\,\text{equiv}_L \text{ (admissible)}}}{\vdash \neg\neg A \Rightarrow A}$$

but this is exactly LEM, not provable in NJ.

Exercise 2.

$$\dfrac{\dfrac{}{\overline{\top} \vdash \top}\,\top_i \qquad \dfrac{\dfrac{\dfrac{}{\top, X \vdash X}\,ax}{\top \vdash X \Rightarrow X}\,\Rightarrow_i}{\top \vdash \overline{\top}}\,\forall_i}{\vdash \overline{\top} \Leftrightarrow \top}\,\wedge_i + \Rightarrow_i$$

$$\dfrac{\dfrac{\dfrac{}{\overline{\bot} \vdash \forall X.X}\,ax}{\overline{\bot} \vdash \bot}\,\forall e(\bot) \qquad \dfrac{\dfrac{}{\bot \vdash \bot}\,ax}{\bot \vdash \overline{\bot}}\,\bot e}{\vdash \overline{\bot} \Leftrightarrow \bot}\,\wedge_i + \Rightarrow_i$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash A \wedge B}\,ax}{\Gamma \vdash A} \quad \dfrac{\dfrac{}{\Gamma \vdash A \wedge B}\,ax}{\Gamma \vdash B}}{\Gamma := A \wedge B, A \Rightarrow B \Rightarrow X \vdash X}\,\Rightarrow e^{\times 2} + ax}{A \wedge B \vdash (A \Rightarrow B \Rightarrow X) \Rightarrow X}\,\Rightarrow_i}{A \wedge B \vdash A \overline{\wedge} B}\,\forall_i \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{A \overline{\wedge} B, A, B \vdash A}\,ax}{A \overline{\wedge} B \vdash A \Rightarrow B \Rightarrow A} \quad \dfrac{\dfrac{}{A \overline{\wedge} B \vdash A \overline{\wedge} B}\,ax}{A \overline{\wedge} B \vdash (A \Rightarrow B \Rightarrow A) \Rightarrow A}}{A \overline{\wedge} B \vdash A}\,\forall E(A) \qquad \Pi}{A \overline{\wedge} B \vdash A \wedge B}}{\vdash A \wedge B \Leftrightarrow A \overline{\wedge} B}\,\wedge_i + \Rightarrow_i$$

We assumed $X \notin FV(A,B)$, (which we can always do as $FV(A,B)$ is finite and we have an infinite number of ty variables).

where

$$\Pi = \dfrac{\dfrac{\dfrac{}{A \overline{\wedge} B, A, B \vdash B}\,ax}{A \overline{\wedge} B \vdash A \Rightarrow B \Rightarrow B} \quad \dfrac{\dfrac{}{A \overline{\wedge} B \vdash A \overline{\wedge} B}\,ax}{A \overline{\wedge} B \vdash (A \Rightarrow B \Rightarrow B) \Rightarrow B}}{A \overline{\wedge} B \vdash B}\,\forall E(B)$$

We have:

$$\dfrac{\Pi \qquad\qquad\qquad\qquad \Pi'}{\vdash A \overline{\vee} B \Leftrightarrow A \vee B}\,\wedge_i + \Rightarrow_i$$

where

$$\Pi = \dfrac{\dfrac{\dfrac{\dfrac{}{A \overline{\vee} B \vdash \forall X. (A \Rightarrow X) \Rightarrow (B \Rightarrow X) \Rightarrow X}\,ax}{A \overline{\vee} B \vdash (A \Rightarrow A \vee B) \Rightarrow (B \Rightarrow A \vee B) \Rightarrow A \vee B}\,\forall E \quad \dfrac{\dfrac{}{A \overline{\vee} B, B \vdash A \vee B}\,\vee_i + ax}{A \overline{\vee} B \vdash B \Rightarrow A \vee B}\,\Rightarrow_i \quad \dfrac{\dfrac{}{A \overline{\vee} B, A \vdash A \vee B}\,\vee_i + ax}{A \overline{\vee} B \vdash A \Rightarrow A \vee B}}{A \overline{\vee} B \vdash A \vee B}$$

and

$$\Pi' = \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{A, A\Rightarrow X, B\Rightarrow X \vdash A\Rightarrow X}\,ax}{A, A\Rightarrow X, B\Rightarrow X \vdash X}\,\Rightarrow e + ax}{A \vdash (A\Rightarrow X)\to(B\Rightarrow X)}\,\Rightarrow i\ \times 2}{A \vdash A \vee B}\,\to i \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{B, A\Rightarrow X, B\Rightarrow X \vdash B\Rightarrow X}\,ax}{B, A\Rightarrow X, B\Rightarrow X \vdash X}\,\Rightarrow e + ax}{B \vdash (A\Rightarrow X)\to(B\Rightarrow X)}\,\Rightarrow i\ \times 2}{B \vdash A \vee B}\,\to i}{A \cup B \vdash A \vee B}\,\vee e + ax + wk$$

And finally, we have

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\neg A, A \vdash \bot}\,\text{contradiction}}{\neg A, A \vdash X}\,\bot e}{\neg A \vdash \neg A}\,\forall i + \Rightarrow i \qquad \dfrac{\dfrac{}{\neg A \vdash \forall X. A\to X}\,ax}{\neg A \vdash \neg A}\,\forall e(\bot)}{\vdash \neg A \iff \neg A}\,\wedge i + \Rightarrow i$$

$$\neg A \equiv A \Rightarrow \bot$$

Exercise 3.

Q1. $list(X) := \forall Y.\ Y \Rightarrow (X \Rightarrow Y \Rightarrow Y) \Rightarrow Y$

with

$nil : \forall X.\ list(X)$

$nil := \Lambda X Y.\ \lambda y f.\ y$

$cons : \forall X.\ X \Rightarrow list(X) \Rightarrow list(X)$

$cons := \Lambda X.\ \lambda x\, l.\ \Lambda Y.\ \lambda y f.\ f\ x\ (l\ Y\ y\ f).$

$fold : \forall X Y.\ Y \Rightarrow (X \Rightarrow Y \Rightarrow Y) \Rightarrow list(X) \Rightarrow Y$

$fold := \Lambda X Y.\ \lambda y\, f\, l.\ \ l\ Y\ y\ f.$

Q2. $\quad map : \forall XY.\ (X \to Y) \Rightarrow (list\ (X) \Rightarrow list(Y))$

$map := \Lambda XY.\ \lambda f\ l.\ fold\ X\ (list\ Y)\ (nil\ Y)$
$\qquad (\lambda x a.\ cons\ Y\ (f\ x)\ a)\ l$

$map\ X\ Y\ f\ \ (cons\ X\ x\ \ (cons\ X\ x'\ (nil\ X)))$

$\to_\beta^* map\ X\ Y\ f\ \ (cons\ X\ x\ \ (\Lambda Y'.\ \lambda y g.\ g\ x'\ (nil\ X\ Y'\ y\ g)))$

$\to_\beta^* map\ X\ Y\ f\ \ (\Lambda Y'.\ \lambda y\ g.\ g\ x\ (g\ x'\ (nil\ X\ Y'\ y\ g)))$

$\to_\beta^* fold\ X\ (list\ Y)\ (nil\ Y)\ (\lambda x a.\ cons\ Y\ (f\ x)\ a)$
$\qquad (\Lambda Y'.\ \lambda y\ g.\ g\ x\ (g\ x'\ y))$

$\to_\beta^* cons\ Y\ (f\ x)\ (cons\ Y\ (f\ x')\ (nil\ Y))$

Q3. $\quad length : \forall X.\ list\ (X) \Rightarrow Nat$

$length := \Lambda X.\ \lambda l.\ fold\ X\ Nat\ zero\ (\lambda x a.\ succ\ a)\ l$

We have that

$$fold\ X\ Y\ y\ f\ \overbrace{(cons\ X\ x_0\ (\dots (cons\ X\ x_{n-1}\ (nil\ X))\dots)}^{\to_\beta^* \Lambda Z \lambda z g.\ g\ x_0 (\dots (g\ x_{n-1}\ (g\ x_{n-1}\ z)\dots)}$$

$$\to_\beta^* f\ x_0\ (f\ x_1\ (\dots (f\ x_{n-1}\ (f\ x_{n-1}\ y))\dots)$$

which we can show by induction on $n \in \mathbb{N}$ (similar to Q2).

So, we conclude that

$$length\ X\ (cons\ X\ x_0\ (\dots (cons\ X\ x_{n-1}\ (nil\ X))\dots)$$

$$\to_\beta^* \underbrace{succ\ (\dots (succ}_{n\ succ}\ zero)) \to_\beta^* \underline{n}$$

$\rightsquigarrow$ Rocq file

Exercise 4.

Q1. $\mathrm{tree}(X) := \forall Y.\,(X \Rightarrow Y) \Rightarrow (Y \Rightarrow Y \Rightarrow Y) \Rightarrow Y.$

$\mathrm{leaf} : \forall X.\, X \Rightarrow \mathrm{tree}(X)$

$\mathrm{leaf} := \Lambda X\, \lambda x.\, \Lambda Y.\, \lambda f\, _.\, f\, x.$

$\mathrm{binNode} : \forall X.\, \mathrm{tree}(X) \Rightarrow \mathrm{tree}(X) \Rightarrow \mathrm{tree}(X)$

$\mathrm{binNode} := \Lambda X.\, \lambda u\, v.\, \Lambda Y.\, \lambda f\, g.\, g\, (u\, Y\, f\, g)\, (v\, Y\, f\, g).$

Q2. $\mathrm{frontier} : \forall X.\, \mathrm{tree}(X) \Rightarrow \mathrm{list}(X)$

$\mathrm{frontier} := \Lambda X.\, u\, (\mathrm{list}\, X)\, (\lambda x.\, \mathrm{cons}\, X\, x\, (\mathrm{nil}\, X))\, (\mathrm{concat}\, X)$

where $\mathrm{concat} : \forall X.\, \mathrm{list}\, X \Rightarrow \mathrm{list}\, X \Rightarrow \mathrm{list}\, X.$

$\mathrm{concat} := \Lambda X.\, \lambda l\, l'.\, \mathrm{fold}\, X\, (\mathrm{list}\, X)\, l'\, (\lambda x\, a.\, \mathrm{cons}\, X\, x\, a)\, l$

Exercise 5.

We have in $N\bar{5}^2$,

$$\dfrac{\dfrac{\dfrac{}{\forall X A, \exists X, \neg A \vdash \exists X\, \neg A}\,ax \qquad \dfrac{\dfrac{\dfrac{}{\forall X A \vdash \forall X A}\,ax}{\forall X A, \vdash A}\,\forall e}{\forall X A, \exists X. \neg A, \neg A \vdash \bot}\,\neg e + ax + wk}{\forall X. A, \exists X. \neg A \vdash \bot}\,\exists e}{\vdash \forall X.\, A \Rightarrow \neg \exists X.\, \neg A}\,\Rightarrow i \times 2$$

but the converse is unprovable in NJ as, if $X \notin FV(A)$, we have

$$\neg\exists X.\neg A \iff \neg\neg A$$

and so we would have to prove LEM in NJ^2 (which is unprovable by conservation).

In NC^2 we do have a proof by replacing $\forall X.A$ with $\neg\neg\forall X.A$ which is admissible in NC^2.

Similarly,

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash A}\,ax \qquad \dfrac{\dfrac{}{\Gamma \vdash \forall X.\neg A}\,ax}{\Gamma \vdash \neg A}\,\forall E}{\Gamma := A, \forall X, \neg A \vdash \bot}\,\neg e}{\exists X.A, \forall X, \neg A \vdash \bot}\,\exists e + ax + wk}{\vdash \exists X A \Rightarrow \neg\forall X.\neg A}\,\Rightarrow i \times 2$$

but the converse is unprovable in NJ^2 for the same reason:

$$\neg\forall X.\neg A \Rightarrow \exists X.A$$

implies when $X \notin FV(A)$ that

$$\underbrace{\neg\neg A \to A}_{\text{LEM.}}$$

In NC^2 it is provable by using LEM to transform it

$$\neg\forall X.\neg A \Rightarrow \neg\neg\exists X.A$$

which, by contra position, is equivalent to

$$\neg\exists X.A \to \forall X.\neg A.$$

and

$$\dfrac{\dfrac{\dfrac{\dfrac{}{A, \neg\exists X.A \vdash A}\,ax}{A, \neg\exists X.A \vdash \exists X.A}\,\exists i}{A, \neg\exists X.A \vdash \bot}\,\Rightarrow e + ax}{\vdash \neg\exists X.A \Rightarrow \forall X.\neg A}\,\Rightarrow i + \forall i + \Rightarrow i$$

Exercise 7.

We define

$$\overline{\exists X}\ A := (\forall Y.\, X \Rightarrow Y) \Rightarrow A$$

it has one constructor:

$$\text{here} : X \Rightarrow \overline{\exists X}.A$$

$$\text{here} := \lambda x f.\ f\ A\ x.$$

and one destructor:

$$\text{use} : (\forall Y.\, X \Rightarrow Y) \Rightarrow \overline{\exists X}.A \Rightarrow A$$

$$\text{use} := \lambda P u.\ u\ P$$

with

$$\text{use}\ P\ (\text{here}\ x) \to_\beta^* \text{here}\ x\ P \to_\beta^* P\ A\ x.$$